

claude-windows / 985cc286-9e1e-4217-87c3-2d418bc4fa9c

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\985cc286-9e1e-4217-87c3-2d418bc4fa9c\subagents\workflows\wf_fc91ae6e-b62\agent-a94ba46b2b5ca188b.jsonl`  
- SHA-256: `1f99ba8d123e2b39eac656d69c598bbe5a964182227c5ce78fb819d04609e9cf`  
- Source modified: `2026-06-14T06:51:41+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_fc91ae6e-b62`  
- session_id: `985cc286-9e1e-4217-87c3-2d418bc4fa9c`
```

Transcript

1. user (2026-06-14T06:50:14.730Z)

In the badminton-highlight-indexer repo (cwd), extract EVERY quantitative result with its source, VERBATIM and accurate (this feeds a paper ? do not round or invent). Read docs/QUALITY_ITERATIONS.md (esp. the "Baselines that drive strategy" section + the Gate-A and fusion entries) and eval_baselines/*.json (heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json, experiment_leaderboard.json). Return: the heuristic/Gen-0/two-stage/wrapper F1 on clean (Mahadevpura) vs multi-court (Kushagra) footage; the Gen-0 LOVO score + floor + sign test; Gate-A promotion stats (?F1, n/6, sign-test p, CI, over-seg ratio); the learned-variant fix (#130) magnitude; the fusion P3 numbers (shuttle-only vs +person: F1, ?F1, 95% CI, sign test wins/losses+p); the fusion P4 incremental-audio numbers (shuttle+person vs +audio: F1, ?F1, CI, sign test); corpus size (n golden videos, total rallies). Put exact numbers in 'numbers' with what each is.

2. assistant (2026-06-14T06:50:17.284Z)

I'll read the files to extract the quantitative results. Let me start with the key documents and JSON baselines.

3. user (2026-06-14T06:50:17.770Z)

```
1      # Rally-detection quality iterations ? living log  
2  
3      **Purpose.** A running, reverse-chronological log of how we evolve rally-detection  
quality: each  
4      iteration's hypothesis, what shipped, the measured (or expected) effect, and the lesson.  
This is the  
5      **live** continuation of the historical trail in  
6      [`archives/quality-iterations/`](archives/quality-iterations/README.md) ? per the  
archive policy  
7      (`CLAUDE.md`), **terminal-state** (DONE/REJECTED) work gets a dated subdir there; this  
file stays at  
8      `docs/` top level and keeps growing, then its entries graduate to the archive when they  
conclude.  
9  
10     **Methodology (non-negotiable).** Every quality change rides the experiment harness  
11     ([EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md)): new cues land **flag-gated, default  
OFF**; they're  
12     graded **offline against golden labels** (per-video + LOVO-honest); promotion to default  
happens **only  
13     on measured lift** through the `run_regression.py` gate; **rollback = flip a config  
flag, never  
14     `git revert`**. **Cues are designed in  
[MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md); scoring  
15     in [EVALUATION.md](EVALUATION.md); the 2-layer model in  
16     [HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md).
```

```

17
18     **Golden corpus = the 6 labelled matches** in `output/human_lovo_manifest.json` (owner-
confirmed
19     2026-06-13: use all 6 ? `adarsh_avi_singles, kushagra_singles, largetest_doubles,
gbaaddy,
20     testlarge_short, mahadevpura_singles`). The model trained on these is what we run
alongside the existing
21     detector on NEW footage for the side-by-side review.
22
23     ---
24
25     ## Baselines that drive strategy (IoU 0.5 vs human golden labels)
26
27     | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
28     |---|---|---|
29     | Heuristic (velocity windowing) | 0.657 | 0.157 |
30     | Gen-0 owned head (LOVO, n=6) | 0.744 | 0.561 |
31     | Two-stage WASB+Gemini refine | 0.83 | ? |
32     | Gemini wrapper (`gemini-flash-latest`) | 0.93 | 0.66 |
33
34     **Reading:** clean footage is commoditized (serve via Gemini); **multi-court drops the
commodity wrapper
35     to 0.66 ? that is the owned model's ship target.** Baselines tracked in
`eval_baselines/`. The owner's
36     observed false positive ? a player **holding/flicking** the shuttle between points
counted as a rally ?
37     is the precision bug the fusion gates target.
38
39     ---
40
41     ## Current track (live) ? newest first
42
43     ### Fusion P3 + P4 MEASURED on the n=6 golden corpus ? person = no lift, audio =
promising (not promotable) *(2026-06-14)*
44     The GPU person-feature extraction finished all 6 golden videos (`fusion_golden.py` ?
`output/*_golden_features.json`, the
45     replayable substrate); the CPU LOVO drivers then measured the lift with **honest stats
(C1 ? bootstrap 95% CI + paired sign
46     test; never a bare mean at n=6)**:
47
48     | Variant (honest LOVO, IoU ? 0.5) | F1 | ?F1 vs prior | 95% CI | sign test | CI clears
0 |
49     |---|---|---|---|---|---|
50     | shuttle-only | 0.307 | ? | ? | ? | ? |
51     | + person (P3) | 0.286 | **?0.021** (?6.7%) | [?0.165, +0.162] | 2W/4L (p=0.688) |
**No** |
52     | + person + audio (P4 incr.) | 0.366 | **+0.079** (+27.7%) | [?0.054, +0.203] | 4W/2L
(p=0.688) | **No** |
53
54     **Read honestly:** neither cue clears the promotion bar (both CIs span 0, both sign
tests n.s.) at n=6 ? **both stay
55     default-OFF** (promote-only-on-lift). **Person** adds noise (no court polygons ?
`players_in_court` counts spectators);
56     per-video it's a wash (testlarge +0.40, gbaaddy ?0.22). **Audio** is the more promising
lead ? +0.079 / +27.7%, wins 4/6
57     (largetest +0.27, mahadevpura +0.27), and shuttle+person+audio (0.366) lands **above**
shuttle-only (0.307) ? but n=6 can't
58     confirm it. **Next:** more matches (raise N until the sign test can resolve ~+0.08), and
measure audio on shuttle-only
59     *directly* (without the person handicap). Recorded in
`eval_baselines/experiment_leaderboard.json`. **Negative/uncertain
60     results are kept on purpose ? this is the honest ablation backbone for the paper aim.**
61
62     ### Acting on the first A/B: Gate-A PROMOTED + learned variant FIXED + GX010125
corroboration *(2026-06-14)*

```

63 ****Gate-A (net-crossings ? 2) ? PROMOTED (first cue promoted via the harness).**** The measured win

64 (?F1 +0.074, ****6/6 golden matches, sign-test p=0.031****, CI [+0.042,+0.104]; over-seg 1.68?1.01 ? the

65 [first-A/B archive](archives/quality-iterations/2026-06-first-ab-experiment/README.md)) clears the honest

66 bar. ****Recommended config:**** `indexing.fusion.min\_crossings = 2` on the fusion path (the eval-only Gate A

67 folded into production, MULTI_SIGNAL_FUSION_PLAN \$3.1). ? The offline win is on the recall-oriented *proposal*

68 candidate set; flipping the ****live served default**** stays owner-gated until a real fusion-path run confirms it

69 (the production fusion segmenter is P2/unverified on real video) ? so this records the promotion + rationale,

70 not a silent default flip. ****Corroborated on real footage:**** offline re-score of GX010125_1080p's *cached*

71 trajectory (no GPU, isolated `output/GX010125\_run/`) ? 45 candidates ? CONTROL 45 rallies vs ****Gate-A 27****;

72 Gate-A dropped ****18 low-crossing windows (~1.7 min)**** of likely held-shuttle/feed FPs (no golden labels for

73 this footage ? divergence, not F1).

74

75 ****Learned variant FIXED ? [#130](https://github.com/avidullu/badminton-highlight-indexer/pull/130).**** The

76 first A/B's learned logistic lost to CONTROL and zeroed `testlarge` ? a fixed-0.5-cutoff artifact. Added

77 additive/default-OFF `Variant.tune\_threshold` (operating point chosen on the ****train**** fold) + `calibrate`

78 (Platt/isotonic, C2). Validated on the n=6 golden set: ****mean F1 0.307 ? 0.423 (+0.116)****, testlarge

79 0.000 ? 0.300, and the learned variant now ****beats CONTROL (0.388)****. Threshold-in-LOVO is the big lever;

80 calibration is available too.

81

82 ****Next:**** a real fusion-path run to confirm Gate-A served-side; per-video output layout

83 (PER_VIDEO_OUTPUT_LAYOUT.md) so each video's artifacts are self-contained.

84

85 ### First end-to-end A/B on the golden corpus (n=6) ? ****ARCHIVED**** *(2026-06-14)*

86 First real use of the harness + C1?C4 helpers, 100% offline (re-scored persisted WASB trajectories, no GPU).

87 ****Gate-A (net-crossings ? 2) is a measured win**** ? ?F1 +0.074, 6/6 matches, sign-test ****p = 0.031****, CI

88 [+0.042, +0.104], over-segmentation (segment-count ratio) ****1.68 ? 1.01****. The window-level 5-feature logistic

89 ****did not clear the bar**** (?0.081, 2W/4L; a threshold/calibration artifact ? *not* the stronger per-frame

90 prototype). The framework held up: ****C1 promoted one variant and refused another; C4 surfaced over-segmentation**

91 tIoU hid.** Full record + tables ?

92 [archives/quality-iterations/2026-06-first-ab-experiment/](archives/quality-iterations/2026-06-first-ab-experiment/README.md).

93 Next: Gate-A is a candidate for the no-regression promotion gate; the logistic failure is the C2 (calibration)

94 + threshold-in-loop task list.

95

96 ### Eval-honesty helpers (course corrections C1?C4) ? additive, default-OFF *(2026-06-14)*

97 ****Why.**** A multi-agent literature review (folded into

98 ANALYZERS_AND_RECONCILERS.md \$13) confirmed the framework's *skeleton* is

99 standard prior art (Temporal Action Localization/Segmentation + late/score-level fusion ? cite, don't claim),

100 and flagged that our ****measurement**** is walking into known small-N traps that would make "promote only on

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101     measured lift" promote noise or hide a real win. Four corrections, shipped as **pure,
additive helpers** (no
102     existing `compare`/LOVO`/metrics.score` behaviour changed ? callers opt in):
103     - **C1 ? never a single mean.** `backend/eval/eval_stats.py`:
`mean_with_ci`/`bootstrap_ci` (deterministic,
104         seeded), `paired_sign_test` (the exact distribution-free "won on enough *videos?"
test the corpus already
105         needs), `paired_delta_ci` (B?A with a CI that must clear 0). At n?6 a lone LOVO mean
is not trustworthy.
106     - **C2 ? calibrate cue confidence.** `backend/eval/score_calibration.py`: Platt +
isotonic (PAVA) calibrators
107         + `brier_score`/`expected_calibration_error`, so a CNN softmax, an audio onset, and an
integer net-crossing
108         count are comparable before the arbiter and the over-confident cue can't swamp the
rest.
109     - **C3 ? honest stacking.** `eval_stats.out_of_fold_predictions`/`grouped_folds` (leave-
one-group-out by
110         match) so a *learned* producer's outputs used as arbiter features are out-of-fold (no
leakage).
111     - **C4 ? over-segmentation-visible metrics.** `backend/eval/segmentation_metrics.py`:
`f1_at_overlaps`
112         (F1@{10,25,50}), `average_precision`/`mean_average_precision` (mAP@tIoU),
`segment_count_ratio`. tIoU alone
113         is blind to a rally split in 3 or two rallies merged ? exactly what the reconciler
exists to fix.
114
115     **Status.** Helpers + 24 unit tests landed (pure stdlib, run in the numpy/GPU-free
lanes); ruff + mypy clean.
116     **Not yet wired into the default harness flow** (wiring `compare`/`run_regression.py` to
report CIs + F1@k is
117         the follow-on, owner-gated). The discipline they encode is the methodology for every
future cue measurement ?
118         see §13/C1?C4 for the citations and the remaining (Tier-2/3) corrections.
119
120     ### Multi-signal fusion P2: the FusionSegmenter *(in progress, 2026-06-14)*
121     **Hypothesis.** Adjudicating shuttle-seeded windows with the net-crossing **Gate A**
(`min_crossings?2`)
122         and a person-occupancy **Gate B** raises **precision** on the held-shuttle / warm-up /
one-sided-feed /
123         spectator false positives ? *without* hurting recall, because the shuttle stays the
primary,
124         camera-invariant seed.
125
126     **Scope ? and an honesty boundary from the archive (read this).** The
127         [2026-06-multivideo-lovo`](archives/quality-iterations/2026-06-multivideo-
lovo/README.md) iteration
128         already smoke-tested **court-region detection as a fix for the *multi-court contrast*
**confound** and
129         found it **low-headroom**: only 12?14% of dead-time shuttle detections fall *outside*
the foreground
130         court, so a perfect court gate lifts contrast to only ~1.6?2.2× (below the ?3.6× "works"
threshold) ? the
131         multi-court problem is **temporal**, and the **learned model**, not a court mask, is its
answer. So Gate B
132         here is **NOT** sold as the multi-court fix. It targets a **different, previously-
**untested** failure mode
133         (held-shuttle/warm-up/one-sided/spectator FPs). Two distinct levers; don't conflate
them.
134
135     **Data support for Gate A.** The same archive shows real rallies on the multi-court
matches contain
136         **?2 net-crossings 97% of the time (median 5)** while a single service feed crosses
~once ? `min_crossings?2`
137         is a principled, data-backed discriminator (it's the held-shuttle owner-bug killer).
138

```

```

139     **Shipping shape.** A NEW `fusion` segmenter, **default OFF** (`default_segmenter`
unchanged); Gate A
140     wired into the live trajectory path behind `indexing.min_crossings` (default 0 = OFF ?
wasb/tracknet
141     byte-identical today); the court mask is an **optional** config polygon (omit ? Gate B
is
142     player-count-only; supply ? masked). Person features (`players_in_court,
two_sided_motion,
143     mean_player_motion, in_court_ratio, shuttle_player_colocation`) persisted into candidate
`stats` for P3's
144     learned fusion. Reuses `PersonDetector` (P1) + `rally_gate` (already existed) ? no
reinvention.
145
146     **Status (2026-06-14): SHIPPED, default-OFF.** `FusionSegmenter`
(`backend/pipeline/segmenters/fusion_hybrid.py`,
147     registry key `fusion_hybrid`) + the pure feature math
(`backend/pipeline/detectors/fusion_features.py`) + two
148     identity hooks on the trajectory engine (`_enrich_candidate_stats`,
`_select_for_handoff`) + `PersonDetector`.
149     detections_per_frame` + the `FusionConfig` block. Built design-first (a
`fusion-p2-design` fan-out workflow), then
150     **adversarially reviewed by a 31-agent workflow (25 findings ? 19 confirmed)**: the
headline confirmation is
151     **NO-REGRESSION verified** ? wasb/tracknet twins persist byte-identical stats and hand
off all windows (pinned by
152     the extended equivalence test); default `FusionSegmenter` == its wasb substrate. The
confirmed low/medium findings
153     were fixed (frame-dim guards, `round()` frame indices, the per-window axis-estimate
redundancy ? a reusable
154     `rally_gate` estimate, honest torch-import docstrings, and 6 new tests closing the
person-detector / Gate-B /
155     substrate / colocation-halo gaps). Suite 441?467 green**; ruff + mypy clean.
156     **Two findings deferred with intent** (reviewer-agreed): person features are computed
eagerly even for
157     Gate-A-pruned windows (a P3 redesign, would change the persisted-superset semantics);
and `fusion.velocity_threshold`
158     must be set equal to the substrate's to A/B against a tuned wasb run (documented in
`FusionConfig`).
159     **Lift still to be measured** offline on the 6 golden videos via
`experiment.compare(CONTROL, fusion)`; the
160     held-shuttle-drop + person-feature values are GPU/owner-side. *(Measured numbers
appended here when run ? no claims before.)*
161
162     ### Person cue extracted to a reusable producer (fusion P1) ?
[#115](https://github.com/avidullu/badminton-highlight-indexer/pull/115), 2026-06-13
163     Lifted the torchvision detector + ByteTrack + velocity windowing out of `yolo_hybrid`
into
164     `backend/pipeline/detectors/person_detector.py::PersonDetector`. **No behavior change**
(proven by the
165     unchanged `process_video` output tests); `yolo_hybrid` now orchestrates only. **Why it's
a quality move:**
166     it makes "add the person cue to fusion" an *add-a-producer*, not a segmenter rewrite.
Suite 440?441 green.
167
168     ### Experiment harness: A/B without rollback risk ?
[#114](https://github.com/avidullu/badminton-highlight-indexer/pull/114), 2026-06-13
169     Shipped `backend/eval/experiment.py` ? the thin variant/experiment layer (the audit's
"-80% already
170     exists" held; pure glue over
`metrics`/`calibration`/`classifier`/`training`/`regression.gt_hash`). Gives
171     `Variant` (+ pinned `CONTROL`), `compare()` (labeled A/B, LOVO-honest),
**`compare_unlabeled()`** (the SxS
172     read on NEW unlabeled footage ? agreed / A-only / B-only), and a JSON experiment
leaderboard. **This is
173     the instrument every iteration is measured on.** 19 tests incl. the no-regression

```

```

invariant (all cues OFF
174     ? byte-identical to control). Suite 421?440 green.
175
176     ---
177
178     ## Archived predecessors (the historical trail)
179     Completed iterations live under [archives/quality-iterations/](archives/quality-
iterations/README.md):
180     - **`2026-06-initial/`** ? seed of the quality phase.
181     - **`2026-06-human-gt-tuning/`** ? first eval vs **human** GT on a partial-label video;
partial-label
182     harness + labeled-window methodology; baseline 0.650 ? tuned 0.742 (*fit-on-self*);
found+fixed a
183     negative-start ingestion bug (collector PR #13).
184     - **`2026-06-multivideo-lovo/`** ? the **n=6 corpus + honest LOVO** ; established the
heuristic is
185     **footage-fragile** (multi-court Kushagra 0.16 / gBaaddy 0.28), the **"contrast"
metric** predicts
186     heuristic F1 monotonically, and a **learned per-frame prototype overtook the heuristic
at n=3 and holds
187     ~+0.10** (n=6: learned 0.583 vs heuristic 0.480), with **AUC 0.83?0.92 decoupled from
contrast**. This
188     is the data-backed green light for the owned model.
189
190     ---
191
192     ## Standing lessons (carry-forward)
193     - **Persist detection once, re-score variants forever.** Expensive/risky = per-frame
detection (GPU/WSL);
194     cheap/swappable = the decision. Most experiments never enter the served pipeline.
195     - **LOVO or it didn't happen.** Score on a held-out *video*, never a held-out window
from the same video.
196     Group by **match**, not file. Sign test needs n ? ~n=10/9-wins for p<0.05.
197     - **The multi-court confound is TEMPORAL, not spatial/loudness** ? three heuristics were
tested and failed
198     for the same reason (foreground spatial gate **?0.04** ; audio energy **AUC ~0.6** ;
court-region
199     separability **12?14% out-of-region**): the dead-time confusion is rally-vs-non-rally
activity in the
200     *same place and frequency band*. Only the learned model's temporal/structural features
crack it (AUC
201     0.83 on Kushagra/gBaaddy). **Don't re-derive these dead ends.**
202     - **Court-masking ? multi-court fix** (see above) ? it's a **precision** tool for off-
court persons and
203     one-sided feeds, a *different* lever; measure its lift, don't assume it.
204     - **Net-crossings are a strong rally signal** ? 97% of real rallies cross ?2x (median
5); single feeds ~1x.
205     Gate A leans on this.
206     - **Small-N feature discipline.** Few, scale/translation-invariant features
(counts/ratios, not raw
207     coords) + a logistic regression, not a net ? raw coords memorize this court/camera.
208     - **Calibration, not representation, is the small-N wall** ? learned probabilities
separate (high AUC) but
209     the operating point doesn't transfer with few training videos. More distinct matches
is the highest-ROI
210     unblock.
211     - **Gate A already existed, just wasn't wired to production** ? the cheapest precision
win is plumbing, not
212     new capability. Always check what the eval drivers already do that the live path
doesn't.
213

```

4. user (2026-06-14T06:50:18.710Z)

```

1      {

```

```

2     "name": "heuristic_velocity_windowing_LOVO",
3     "description": "The honest cross-video floor the Gen-0 learned model must beat:
calibrated velocity-windowing heuristic, leave-one-video-out over the n=6 owned golden corpus.",
4     "computed": "2026-06-10, reproduced via: python -m backend.eval.calibrate_local
--manifest output/human_lovo_manifest.json --metric f1 (matches docs/archives/quality-
iterations/2026-06-multivideo-lovo)",
5     "mean_f1": 0.480,
6     "std": 0.242,
7     "per_video": {
8         "adarsh_avi_singles": 0.751,
9         "kushagra_singles": 0.157,
10        "largetest_doubles": 0.737,
11        "gbaaddy": 0.309,
12        "testlarge_short": 0.267,
13        "mahadevpura_singles": 0.657
14    }
15 }
16

```

5. user (2026-06-14T06:50:19.182Z)

```

1     {
2         "name": "gemini_wrapper_baseline",
3         "description": "Commodity-VLM wrapper baseline: the full pipeline's `gemini` segmenter
scored vs human golden labels at IoU 0.5. The commoditization/moat reference the owned model
must be judged against ? on multi-court footage the wrapper drops 0.93 -> 0.66, so the owned
model's target is the multi-court number, not the clean-footage one.",
4         "model": "gemini-flash-latest (resolved 2026-06-10; 3.5-flash era)",
5         "conditions": "serial uploads (ai_max_workers=1), chunks re-encoded to 1280px w/ audio
(ai_chunk_scale_width), 6-attempt exponential generate retry; min_segment_duration=3.0 filter
active (rejects real ~2.5s rallies ? see notes)",
6         "per_video": {
7             "mahadevpura_singles": {
8                 "f1": 0.93, "precision": 1.00, "recall": 0.87, "mean_iou": 0.80,
9                 "boundary_err_s": 0.7, "n_gt": 31, "n_pred": 27,
10                "notes": "3 of 4 FNs were detected by Gemini but rejected by
min_segment_duration=3.0 (2.5s rallies). Chunk 6 (unlabeled tail) lost to 503s ? no effect on
the labeled window."
11            },
12            "kushagra_singles": {
13                "f1": 0.66, "precision": 0.69, "recall": 0.62, "mean_iou": 0.73,
14                "boundary_err_s": 1.2, "n_gt": 32, "n_pred": 29,
15                "notes": "Multi-court (contrast 1.4x). Full 7/7 chunk coverage, audio intact. FPS
cluster as background-game pickups + one 53s dead-time merge ? the exact confound the contrast
metric predicts."
16            }
17        },
18        "comparison_at_time_of_run": {
19            "heuristic_lovo": {"mahadevpura_singles": 0.657, "kushagra_singles": 0.157},
20            "gen0_learned_lovo": {"mahadevpura_singles": 0.744, "kushagra_singles": 0.561}
21        },
22        "reliability_note": "gemini-flash-latest served 503 capacity waves throughout (one
chunk needed a ~5min backoff); a sole-provider dependency is operationally flaky ? the provider-
registry fallback design is load-bearing.",
23        "cost_note": "well under $1 for both videos at flash pricing (serial, ~14 chunk
requests total)."
24    }
25

```

6. user (2026-06-14T06:50:19.696Z)

```

1     [
2         {
3             "timestamp": "2026-06-14T06:23:51.507393+00:00",
4             "iou": 0.5,

```

```

5      "label_set_hash": "50d98ffbc370",
6      "num_videos": 6,
7      "variant_a": {
8          "name": "shuttle_only",
9          "spec_hash": "47c867c36f2f",
10         "aggregate": {
11             "precision": 0.31807603780927995,
12             "recall": 0.3101947096042718,
13             "f1": 0.3068611057861412,
14             "mean_iou": 0.5791831889148886
15         }
16     },
17     "variant_b": {
18         "name": "shuttle_person_fusion",
19         "spec_hash": "253f4709b4c4",
20         "aggregate": {
21             "precision": 0.3617228001510626,
22             "recall": 0.2708939086387243,
23             "f1": 0.2863398846047612,
24             "mean_iou": 0.5690111885714347
25         }
26     },
27     "delta": {
28         "precision": 0.04364676234178266,
29         "recall": -0.0393008009655475,
30         "f1": -0.02052122118138,
31         "mean_iou": -0.01017200034345389
32     },
33     "honest_delta_f1": {
34         "mean_delta": -0.02052122118137999,
35         "ci_low": -0.16462806145542527,
36         "ci_high": 0.16224446866060574,
37         "n": 6.0,
38         "ci_excludes_zero": false,
39         "sign_test": {
40             "wins": 2.0,
41             "losses": 4.0,
42             "ties": 0.0,
43             "n_effective": 6.0,
44             "p_value": 0.6875
45         }
46     }
47 },
48 {
49     "timestamp": "2026-06-14T06:24:56.487711+00:00",
50     "iou": 0.5,
51     "label_set_hash": "50d98ffbc370",
52     "num_videos": 6,
53     "variant_a": {
54         "name": "shuttle_person",
55         "spec_hash": "ee89f1e41604",
56         "aggregate": {
57             "precision": 0.3617228001510626,
58             "recall": 0.2708939086387243,
59             "f1": 0.2863398846047612,
60             "mean_iou": 0.5690111885714347
61         }
62     },
63     "variant_b": {
64         "name": "shuttle_person_audio",
65         "spec_hash": "d68c39eba87c",
66         "aggregate": {
67             "precision": 0.4530798177857001,
68             "recall": 0.3408138120839734,
69             "f1": 0.3655826552797436,

```



```

70         "mean_iou": 0.7074802852140344
71     }
72 },
73     "delta": {
74         "precision": 0.0913570176346375,
75         "recall": 0.06991990344524907,
76         "f1": 0.07924277067498242,
77         "mean_iou": 0.13846909664259965
78     },
79     "honest_delta_f1": {
80         "mean_delta": 0.07924277067498244,
81         "ci_low": -0.053508315003560535,
82         "ci_high": 0.2029045239922802,
83         "n": 6.0,
84         "ci_excludes_zero": false,
85         "sign_test": {
86             "wins": 4.0,
87             "losses": 2.0,
88             "ties": 0.0,
89             "n_effective": 6.0,
90             "p_value": 0.6875
91         }
92     }
93 }
94 ]

```

7. assistant (2026-06-14T06:50:24.455Z)

I have all the data I need from the three JSON files and QUALITY_ITERATIONS.md. I have cross-references for every number. Let me compile the verbatim extraction.

All numbers are sourced directly. Let me note one important reconciliation: the `experiment\_leaderboard.json` carries fusion F1 values to full float precision, while QUALITY_ITERATIONS.md rounds them to 3 decimals (e.g., 0.307, 0.286, 0.366). I'll report both the doc-displayed and the raw-JSON precision so the paper has exact values.

8. user (2026-06-14T06:51:31.175Z)

Structured output provided successfully

9. assistant (2026-06-14T06:51:41.350Z)

I have extracted every quantitative result verbatim from the four sources and returned it via StructuredOutput. Key highlights captured with exact precision:

- **Baselines table** (clean Mahadevpura / multi-court Kushagra): Heuristic 0.657 / 0.157; Gen-0 owned head 0.744 / 0.561; Two-stage WASB+Gemini 0.83 / ?; Gemini wrapper 0.93 / 0.66.
- **Gen-0 LOVO**: learned mean F1 0.583 vs heuristic floor 0.480 (std 0.242), n=6; sign-test n-rule documented.
- **Gate-A promotion**: ?F1 +0.074, 6/6, p=0.031, CI [+0.042,+0.104], over-seg 1.68?1.01.
- **#130 learned-variant fix**: mean F1 0.307?0.423 (+0.116), testlarge 0.000?0.300, beats CONTROL 0.388.
- **Fusion P3 (+person)**: ?F1 ?0.021, CI [?0.165,+0.162], 2W/4L p=0.6875 ? with the full-precision JSON floats.
- **Fusion P4 (+audio)**: ?F1 +0.079, CI [?0.054,+0.203], 4W/2L p=0.6875 ? with full-precision floats.
- **Corpus**: n=6 golden videos. I flagged that **total rally count** is not aggregated anywhere** in these files ? only per-video n_gt for two videos (Mahadevpura 31, Kushagra 32) exist, so I did not invent a corpus-wide total.

I also flagged two precision/transcription notes for the paper: the doc rounds fusion F1/?F1 while the leaderboard JSON carries full floats (use the JSON), and the doc writes the sign-test p as 0.688 where the JSON stores 0.6875.